

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Topic 3.2 - Noise pollution

A real classroom situation

A school in Karachi sits where two main roads meet. By afternoon, the Class 6 teacher has stopped shouting over the traffic; she uses a hand signal and the pupils copy notes from the board in silence. A visitor asks whether the noise bothers them. "Bother?" says the teacher. "We have stopped noticing it. That is what worries me."

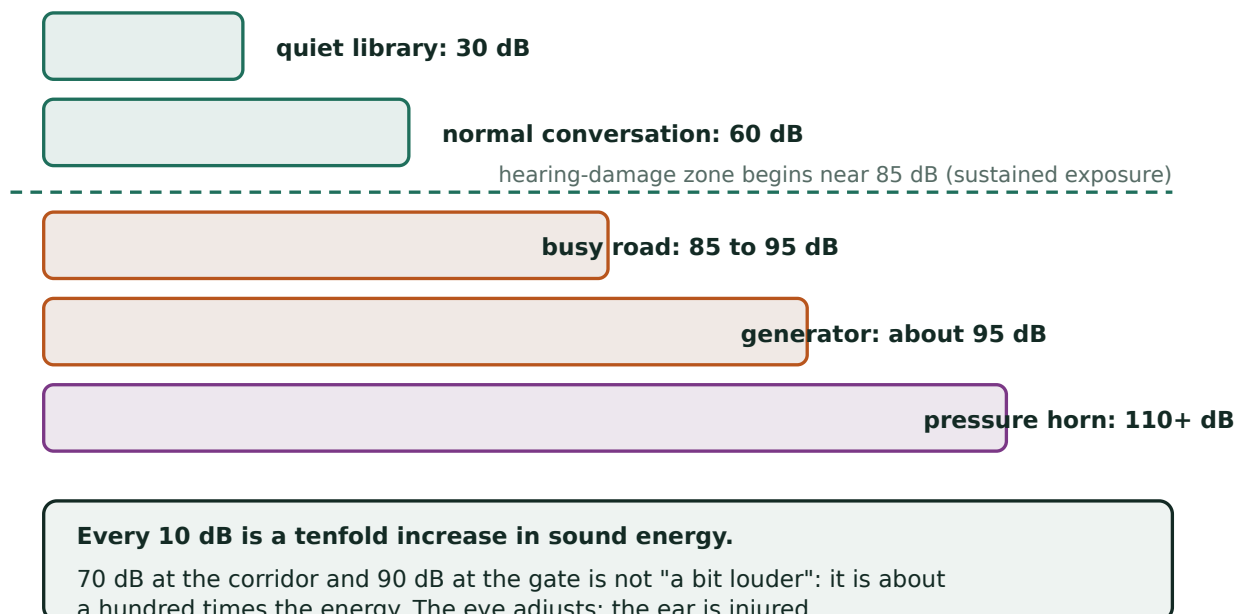
She is right to worry. Not noticing is not the same as not being harmed. This topic is about the pollution people cannot see, often stop noticing, and are still injured by.

Explanation

Noise pollution: sources, human health impacts and control measures

Sound becomes noise pollution when it is loud enough, or persistent enough, to harm health or wellbeing. Loudness is measured in decibels, dB, on a logarithmic scale: every 10 dB is a tenfold increase in sound energy. A whisper is about 30 dB, normal conversation 60, a busy Karachi road 85 to 95, a pressure horn at the window over 110. Sustained exposure above about 85 dB damages hearing; a single very loud event can do it instantly.

The decibel ladder: how loud is it?



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The sources in Pakistan are the ones you can list from any window: road traffic, pressure horns, construction and generators, loudspeakers, workshops and industry. The health impacts go beyond the ear. Hearing loss is the clearest: it is gradual, painless and permanent. The World Health Organization reports that about 430 million people, including 34 million children, already need rehabilitation for disabling hearing loss, and names loud noise exposure among its causes (WHO, 2026). But sustained noise also raises blood pressure and stress hormones, disturbs sleep, and, in classrooms, measurably slows learning: a teacher shouting over traffic is spending her voice and the pupils' attention at the same time.

Noise sources: impact, exposure, control

Source	Health impact in one line	Who is exposed	One control
Road traffic	stress, blood pressure, disturbed sleep	everyone on the street and beside it	enforced horn bans; smooth traffic flow
Construction	hearing loss for workers; disturbed lessons nearby	workers, pupils of the neighbouring	time limits near schools; barriers
Generators	sleep loss, stress in load-shedding hours	households and neighbours	sound boxes; siting; maintenance
Loudspeakers	sleep disturbance; conflict, stress	whole neighbourhoods	volume limits and agreed hours
Workshops	gradual permanent hearing loss	workers, apprentices, the lane	ear protection; machine maintenance

Controls act at the source, in the pathway (distance, barriers, siting), and at the receptor.

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Control measures, once again, sit at the three links. At the source: silencers, horn bans that are actually enforced, quieter machinery, maintenance (a loose part is a loud part). In the pathway: distance, barriers, green belts, and siting, the decision that keeps schools away from main roads, which is the cheapest control of all. At the receptor: ear protection for workers, quiet rooms and, for a school, double glazing and classroom scheduling that keeps the sensitive hours away from peak traffic.

A worked example. The school at the crossroads measures its courtyard with a free sound meter app: 72 dB in the morning, 88 dB at the evening rush. The 16 dB difference is not "a bit louder"; it is about forty times the sound energy. The school moves its listening exercises to the morning, plants a hedge along the wall, and writes to the union council about the pressure horns. Three links, three actions, two of them free.

Common misconception. "Noise is annoying but not a health hazard." Annoyance is the smallest of its effects. Hearing loss is permanent, cardiovascular effects are documented, and children in noisy schools lose learning time daily. Noise is pollution by the same standard as smoke: a physical agent that harms at a dose.

Activity: Measure three places

Work in pairs. Using a free sound meter app on one phone, measure three spots for one minute each: the classroom in silence, the corridor at break, and the nearest road at dismissal time. Record the three numbers, compare them with the ladder figure, and write one sentence per spot naming who is exposed and for how long. Allow 20 minutes, including walking.

Check your understanding

1. What does it mean that decibels are logarithmic? Use a 10 dB example. *(Understand)*
2. Name four health impacts of noise pollution, only one of which is hearing loss. *(Remember)*
3. The corridor measures 70 dB and the road 90 dB. Explain why this is not "twice as loud" in energy terms. *(Analyze)*
4. Your school is beside a main road. Name one control at each of the three links, with the cheapest first. *(Apply)*

Summary

Noise pollution is sound at levels that harm health: hearing loss that is gradual and permanent, plus raised blood pressure, stress, disturbed sleep and lost learning. Decibels are logarithmic, so small number jumps are large energy jumps, and the damage zone begins around 85 dB. Sources are traffic, horns, construction, generators and loudspeakers. Controls act at the source, in the pathway, where distance and siting are cheapest, and at the receptor.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can explain why 90 dB is far more than "a bit louder" than 70 dB.
- ☐ I can name four health impacts of noise beyond annoyance.
- ☐ I can measure sound with an app and place the reading on the ladder.

☐ I can name one noise control at each link of the chain.

Try this at your practicum school

With permission, take three one-minute readings at the school: a quiet classroom, the courtyard at break, and the gate at dismissal. Note the time of day for each. Bring back the three numbers and one recommendation you could make at a staff meeting, with the cheapest option first.

Summative task

Write 150 to 180 words on the following.

The head teacher says, "We cannot move the road, so there is nothing to do about the noise." Using the three links and the evidence of health impacts, write the reply you would give, ending with the two cheapest actions the school could take this term.

Mini-rubric

Criterion	What a strong response does
Links used (Analyze)	Shows that the road is only the source, not the whole problem
Health evidence (Understand)	Names impacts beyond annoyance, correctly
Actions chosen (Apply)	Two feasible, cheap, this-term actions, not a bypass petition
Clarity and register	Plain sentences for a head teacher; within the word count

Further reading

- World Health Organization. (2026). *Deafness and hearing loss* (fact sheet, 3 Mar 2026): <https://www.who.int/news-room/fact-sheets/detail/deafness-and-hearing-loss>
- Phoon, W. O., & Chen, P. C. Y. *Environmental Science in South East Asia*. (guide's listing; unresolvable as printed, cited by reference only; flagged per D-2026-0001)

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